DIRECT RECONSTRUCTION OF ANISOTROPIC SELF-ADJOINT INCLUSIONS IN THE CALDERÓN PROBLEM

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Abstract. We extend the monotonicity method for direct exact reconstruction of inclusions in the partial data Calderón problem, to the case of anisotropic conductivities in any spatial dimension $d \geq 2$. Specifically, from a local Neumann-to-Dirichlet map, we give reconstruction methods of inclusions based on unknown anisotropic self-adjoint perturbations to a known anisotropic conductivity coefficient. A main assumption is a definiteness condition for the perturbations near the outer inclusion boundaries. This additionally provides new insights into the non-uniqueness issues of the anisotropic Calderón problem.

Keywords: Anisotropic Calderón problem, inclusion detection, monotonicity method.

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Contents

1. Introduction

The anisotropic Calderón problem, the inverse problem of determining an elliptic matrix-valued conductivity coefficient A from boundary measurements of the conductivity equation

$$-\nabla \cdot (A\nabla u) = 0 \text{ in } \Omega,$$

is of major interest in the inverse problems community. Unlike the isotropic (scalar-valued) case, the anisotropic Calderón problem suffers from significant non-uniqueness issues. Indeed if boundary measurements are taken on $\Gamma \subseteq \partial\Omega$, for any H^1-diffeomorphism $\Phi\colon \overline{\Omega} \to \overline{\Omega}$ for which $\Phi|_\Gamma$ is the identity, the conductivity coefficient defined via the push-forward

$$\widetilde{A} = \frac{\mathrm{D}\Phi\, A\, \mathrm{D}\Phi^{\mathrm{T}}}{|\det \mathrm{D}\Phi|} \circ \Phi^{-1} \tag{1.1}$$

produces the same local Neumann-to-Dirichlet (ND) map as the coefficient A. For $\Gamma = \partial\Omega$ and with spatial dimension $d = 2$, the obstructions to uniqueness are characterized by (1.1) for real-valued symmetric L^∞-conductivities [4, 37]. In spatial dimension $d \geq 3$, the full extent of the non-uniqueness issues (beyond real-analytic coefficients) remains one of the most important unanswered questions in the inverse problems community [32, 39]. In a fractional Calderón problem, using the stronger non-local properties, the obstructions to uniqueness are characterized for $d \geq 2$ [9]. Despite knowledge on the possible obstructions, the non-uniqueness has prevented development of practically useful reconstruction methods. Unique recovery is only guaranteed with very strong restrictions on the considered class of coefficients [2, 3, 10].

In this paper we study the exact reconstruction of anisotropic inclusions in any spatial dimension $d \geq 2$. We reconstruct the outer shape $D^\bullet$ of

$$D = \operatorname{supp}(A_D - A_0),$$

from knowledge of A_0 and the local ND map $\Lambda(A_D)$ associated with the unknown conductivity A_D. Here $D^\bullet$ is the smallest closed set with connected complement such that $D \subseteq D^\bullet$. We give direct reconstruction methods (Theorems 3.6 and 3.7) based on the monotonicity of ND maps. The main assumptions are self-adjointness of the coefficients, that A_0 satisfies a unique continuation principle, and that $A_D - A_0$ satisfies a local uniform definiteness condition inside D near $\partial D^\bullet$; see Assumption 3.4 for the precise statement. In particular, inside D we do not assume more regularity than L^∞ for A_D, and $A_D - A_0$ can even be indefinite at a positive distance away from $\partial D^\bullet$.

As an indirect consequence: if such a perturbation $A_D - A_0$, to a given "background conductivity" A_0, is known to satisfy the definiteness condition near $\partial D^\bullet$, this is enough to ensure that all coefficient-transformations (1.1), compatible with this condition, satisfy

$$\operatorname{supp}(\widetilde{A}_D - A_0)^\bullet = D^\bullet.$$

Moreover, the inclusion detection problem is also unaffected by any other potential forms of non-uniqueness that could arise from the anisotropic Calderón problem when $d \geq 3$. Hence, for inclusion detection, practical reconstruction is sensible in a very high generality compared to reconstructing an anisotropic conductivity.

To give some intuition on why this is the case, let

$$\widetilde{D} = \Phi(D)$$

be the transformed inclusion under Φ. If $\widetilde{D}^\bullet \neq D^\bullet$ then Φ will fail to be the identity in an open set intersecting $\partial D^\bullet$ (the part where it is deformed by Φ), which also changes the background conductivity in $\widetilde{A}_D$ in that part of the domain via (1.1). From our results we conclude in this situation, that for

$$\widehat{D} = \operatorname{supp}(\widetilde{A}_D - A_0),$$

if $\widehat{D}^\bullet \neq D^\bullet$, the definiteness condition cannot be satisfied inside $\widehat{D}$ near $\partial\widehat{D}^\bullet$. In particular, it may fail in part of the domain where the background conductivity in $\widetilde{A}_D$ is modified.

In the context of Helmholtz scattering, see also the uniqueness results in [21, 6] for inclusion detection with some anisotropic coefficients.

1.1. **The monotonicity method and related results.** The reconstruction methods we present are based on the monotonicity of ND maps via the following operator inequalities (Proposition 5.5):

$$\int_\Omega (A_2 - A_1)\nabla u_2 \cdot \overline{\nabla u_2}\,\mathrm{d}x \leq \langle f, (\Lambda_1 - \Lambda_2) f\rangle \leq \int_\Omega A_2 A_1^{-1}(A_2 - A_1)\nabla u_2 \cdot \overline{\nabla u_2}\,\mathrm{d}x,$$

where $\Lambda_j = \Lambda(A_j)$ and u_2 is the electric potential for conductivity A_2 and Neumann condition f. The definiteness properties of $A_2 A_1^{-1}(A_2 - A_1)$ turn out to be the same as for $A_2 - A_1$ (Proposition 5.6). The main idea in the *outer approach* of the monotonicity method, given in our main results Theorems 3.6 and 3.7, is to construct test-operators that use the monotonicity to check if a given test-set C fully contains $D^\bullet$ or not. This leads to an *if and only if* statement related to operator inequalities with the test-operators and the ND map for the unknown perturbed conductivity. In Theorem 3.7 the test-operators are linearized, which is beneficial for numerical implementation.

The proofs rely on the theory of localized potentials [20], for the existence of certain localizing solutions to the conductivity equation, based on a unique continuation principle. We generalize the result on localized potentials to anisotropic (even non-self-adjoint) coefficients in Theorem 6.3.

Under much stronger assumptions, we also give an *inner approach* of the monotonicity method in Theorems 4.2 and 4.3, which checks if a given open set B is contained in $D^\bullet$.

The monotonicity method is well-established for *isotropic* conductivities, and for other similar inverse coefficient problems. We mention some of the results here, with a focus on the Calderón problem. In [38] they use the monotonicity to give bounds on inclusions, and later [26, 29] prove that it is an exact reconstruction method for finite perturbations to the background conductivity, using the theory of localized potentials. It also includes the case with both positive and negative perturbations, if lower and upper bounds are known for the perturbed conductivity. This is expanded upon in [7], and now the perturbed conductivity can simultaneously have parts with finite positive and negative perturbations, as well as extreme parts that are perfectly insulating and perfectly conducting. The need for lower and upper bounds is also removed. Finally, in [18] the method is generalized to reconstruct collections of Lipschitz cracks.

The method has also been used in other specialized directions: degenerate and singular perturbations based on A_2-Muckenhoupt weights [15], inclusions in a fractional Calderón problem [27, 28], inclusions in a p-Laplace equation [5], for nonlinear materials [36], and for local uniqueness in an inverse coefficient problem for a non-resonant Schrödinger equation [25]. There are also rigorous connections to practically relevant electrode models [16, 17, 22, 30].

Certain reconstruction methods for the partial data isotropic Calderón problem also rely on the same type of methodology. For piecewise constant layered conductivities, there is an exact reconstruction method for both conductivity coefficient and the piecewise constant partition [11, 12]. In a finite-dimensional settting (where the partition is known), a general piecewise constant conductivity is reconstructed in [13], and likewise via a reformulation to a convex semidefinite optimization problem in [23].

1.2. **Remarks on notation.** All considered vector spaces are complex; this includes the classes of conductivity coefficients. At all occurrences of using support/supremum/infimum, we make use of the *essential* support/supremum/infimum.

We use the convention that inner products on complex Hilbert spaces are linear in the first entry and anti-linear in the second. We denote the Euclidean inner product as $z_1 \cdot \overline{z_2}$ for $z_1, z_2 \in \mathbb{C}^d$, in particular the "dot" is bilinear. The Euclidean norm is denoted $|\cdot|$.

For self-adjoint operators $A, B \in \mathscr{L}(H)$ for a Hilbert space H, we write $A \geq B$ when $A - B$ is positive semidefinite; this is the Loewner ordering of such operators. Moreover, for A being a positive definite operator, $A^{1/2}$ denotes its unique positive definite square root.

2. The forward problem

Let Ω be a bounded Lipschitz domain in $\mathbb{R}^d$, for $d \in \mathbb{N} \setminus \{1\}$, with connected complement. For a bounded matrix-valued function $A \in L^\infty(\Omega)^{d\times d}$, we use the norm

$$\|A\|_* = \sup_{x\in\Omega} \|A(x)\|_2,$$

where $\|\cdot\|_2$ is the usual spectral norm for matrices, i.e. the operator norm on Euclidean space. Hence, we have

$$|A(x)\xi| \leq \|A\|_* |\xi|$$

for all $\xi \in \mathbb{C}^d$ and a.e. $x \in \Omega$.

We use the notation

$$A^{\mathrm{R}} = \frac{1}{2}\big(A + A^*\big) \quad \text{and} \quad A^{\mathrm{I}} = \frac{1}{2\mathrm{i}}\big(A - A^*\big),$$

such that

$$A = A^{\mathrm{R}} + \mathrm{i}A^{\mathrm{I}},$$

where we note that A^{R} and A^{I} are self-adjoint as matrices, i.e. Hermitian at every point in Ω. This gives rise to the following general class of anisotropic coefficients:

$$\mathcal{H}(\Omega) = \{\, A \in L^\infty(\Omega)^{d\times d} \mid \exists c > 0\colon A^{\mathrm{R}} \geq cI \text{ in the Loewner order in } \Omega \,\}.$$

The condition on A, the uniform positive definiteness of A^{R}, can also explicitly be written:

$$\inf_{x\in\Omega} \mathrm{Re}\big(A(x)\xi \cdot \overline{\xi}\big) = \inf_{x\in\Omega} A^{\mathrm{R}}(x)\xi \cdot \overline{\xi} \geq c|\xi|^2$$

for all $\xi \in \mathbb{C}^d$; there are other classes of strongly elliptic coefficients (when restricting to $\xi \in \mathbb{R}^d$) that we do not consider here, see e.g. [34, chapter 4]. We also have the subset where A is self-adjoint, i.e. when $A = A^{\mathrm{R}}$:

$$\mathcal{H}_{\mathrm{SA}}(\Omega) = \{\, A \in \mathcal{H}(\Omega) \mid A^{\mathrm{I}} = 0 \,\}.$$

Consider the partial data anisotropic conductivity problem for $A \in \mathcal{H}(\Omega)$:

$$-\nabla \cdot (A\nabla u) = 0 \text{ in } \Omega, \qquad \nu \cdot (A\nabla u) = \begin{cases} f & \text{on } \Gamma, \\ 0 & \text{on } \partial\Omega \setminus \Gamma. \end{cases} \tag{2.1}$$

Here ν is the outer unit normal to Ω, $\Gamma \subseteq \partial\Omega$ is a non-empty open boundary piece, and the mean-free current density f belongs to

$$L^2_\diamond(\Gamma) = \{\, f \in L^2(\Gamma) \mid \langle f, 1\rangle = 0 \,\}.$$

We denote by $\langle \cdot\, , \cdot \rangle$ the standard inner product on $L^2(\Gamma)$, and $\|\cdot\|$ is the associated norm. We likewise denote

$$H^1_\diamond(\Omega) = \{\, u \in H^1(\Omega) \mid \langle u|_\Gamma, 1\rangle = 0 \,\}.$$

A Poincaré inequality, related to the mean-free condition in $H^1_\diamond(\Omega)$, enables that

$$\|v\|_{H^1_\diamond(\Omega)} = \Big(\int_\Omega |\nabla v|^2 \,\mathrm{d}x\Big)^{1/2}$$

becomes a norm on $H^1_\diamond(\Omega)$, equivalent with the usual H^1-norm. Moreover, the following continuity and coercivity estimates hold for all $w, v \in H^1_\diamond(\Omega)$:

$$\Big|\int_\Omega A\nabla w \cdot \overline{\nabla v}\,\mathrm{d}x\Big| \le \|A\|_* \|w\|_{H^1_\diamond(\Omega)} \|v\|_{H^1_\diamond(\Omega)},$$

$$\Big|\int_\Omega A\nabla v \cdot \overline{\nabla v}\,\mathrm{d}x\Big| \ge \operatorname{Re}\int_\Omega A\nabla v \cdot \overline{\nabla v}\,\mathrm{d}x \ge c\|v\|^2_{H^1_\diamond(\Omega)}.$$

By the Lax–Milgram lemma, there exists a unique solution $u = u^A_f \in H^1_\diamond(\Omega)$ to the corresponding weak problem associated to (2.1):

$$\int_\Omega A\nabla u \cdot \overline{\nabla v}\,\mathrm{d}x = \langle f, v\rangle \tag{2.2}$$

for all $v \in H^1_\diamond(\Omega)$. We let $\Lambda(A) \in \mathscr{L}(L^2_\diamond(\Gamma))$ denote the associated local ND map,

$$\Lambda(A)f = u^A_f|_\Gamma,$$

and the forward problem is the nonlinear mapping $\Lambda\colon \mathcal{H}(\Omega) \to \mathscr{L}(L^2_\diamond(\Gamma))$. From (2.2) it holds that for any $A_1, A_2 \in \mathcal{H}(\Omega)$ and $f, g \in L^2_\diamond(\Gamma)$, we have

$$\langle f, \Lambda(A_1)g\rangle = \int_\Omega A_2\nabla u^{A_2}_f \cdot \overline{\nabla u^{A_1}_g}\,\mathrm{d}x. \tag{2.3}$$

Thus the adjoint operator satisfies

$$\langle f, \Lambda(A_1)^* g\rangle = \overline{\langle g, \Lambda(A_1) f\rangle} = \int_\Omega \overline{A_2\nabla u^{A_2}_g} \cdot \nabla u^{A_1}_f \,\mathrm{d}x.$$

Of particular interest are the quadratic forms

$$\langle f, \Lambda(A)f\rangle = \int_\Omega A\nabla u^A_f \cdot \overline{\nabla u^A_f}\,\mathrm{d}x \quad\text{and}\quad \langle f, \Lambda(A)^* f\rangle = \int_\Omega A^*\nabla u^A_f \cdot \overline{\nabla u^A_f}\,\mathrm{d}x.$$

Specifically,

$$\Lambda(A)^* = \Lambda(A^*),$$

meaning that $\Lambda(A)$ is self-adjoint if $A \in \mathcal{H}_{\mathrm{SA}}(\Omega)$. We also write

$$\Lambda^{\mathrm{R}}(A) = \frac{1}{2}\big(\Lambda(A) + \Lambda(A)^*\big) \quad\text{and}\quad \Lambda^{\mathrm{I}}(A) = \frac{1}{2\mathrm{i}}\big(\Lambda(A) - \Lambda(A)^*\big),$$

so for the forward problem we have

$$\Lambda = \Lambda^{\mathrm{R}} + \mathrm{i}\Lambda^{\mathrm{I}},$$

where $\Lambda^{\mathrm{R}}(A)$ and $\Lambda^{\mathrm{I}}(A)$ are self-adjoint operators. In terms of the quadratic forms we thus have

$$\langle f, \Lambda^{\mathrm{R}}(A) f\rangle = \mathrm{Re}\langle f, \Lambda(A) f\rangle = \int_\Omega A^{\mathrm{R}} \nabla u_f^A \cdot \overline{\nabla u_f^A}\,\mathrm{d}x = \int_\Omega |(A^{\mathrm{R}})^{1/2}\nabla u_f^A|^2\,\mathrm{d}x, \tag{2.4}$$

$$\langle f, \Lambda^{\mathrm{I}}(A) f\rangle = -\,\mathrm{Im}\langle f, \Lambda(A) f\rangle = -\int_\Omega A^{\mathrm{I}} \nabla u_f^A \cdot \overline{\nabla u_f^A}\,\mathrm{d}x. \tag{2.5}$$

As one would expect, when multiplying with a complex scalar $\kappa \in \mathbb{C}$, we get

$$(\kappa\Lambda)^{\mathrm{R}} = \kappa^{\mathrm{R}}\Lambda^{\mathrm{R}} - \kappa^{\mathrm{I}}\Lambda^{\mathrm{I}} \quad \text{and} \quad (\kappa\Lambda)^{\mathrm{I}} = \kappa^{\mathrm{I}}\Lambda^{\mathrm{R}} + \kappa^{\mathrm{R}}\Lambda^{\mathrm{I}}.$$

3. Outer reconstruction of general anisotropic inclusions

In the following we will assume that a so-called "background conductivity" $A_0 \in \mathcal{H}_{\mathrm{SA}}(\Omega)$ is known. For an unknown $A_D \in \mathcal{H}_{\mathrm{SA}}(\Omega)$, we call

$$D = \mathrm{supp}(A_D - A_0)$$

the *inclusions*, noting that D may have several connected components. Our reconstruction methods will relate to the so-called outer shape of D.

Definition 3.1. The outer shape $D^\bullet$ of D, is the smallest closed set with connected complement such that $D \subseteq D^\bullet$.

One of the key assumptions we will need, required for the existence of certain localized solutions, is that the background conductivity A_0 satisfies a unique continuation principle.

Definition 3.2. Suppose $U \subseteq \overline{\Omega}$ is a relatively open and connected set that intersects Γ. We say that $A \in \mathcal{H}(\Omega)$ satisfies the weak *unique continuation principle* (UCP) in U for the conductivity equation, provided that only the trivial solution of

$$-\nabla \cdot (A\nabla u) = 0 \text{ in } U^\circ$$

can be identically zero in a non-empty open subset of U, and only the trivial solution has vanishing Cauchy data, u and $\nu \cdot (A\nabla u)$, on a non-empty open part of $\partial U \cap \Gamma$. If $U = \overline{\Omega}$ we just say that A satisfies the UCP.

Remark 3.3. In spatial dimension $d = 2$, the UCP holds for general real-valued $A \in \mathcal{H}_{\mathrm{SA}}(\Omega)$ (and even in some non-self-adjoint cases), see [1].

In spatial dimension $d \geq 3$, the UCP holds for real-valued Lipschitz continuous coefficients $A \in \mathcal{H}_{\mathrm{SA}}(\Omega)$, see [19], and with counterexamples to the UCP for α-Hölder continuous coefficients with $\alpha \in (0,1)$, see [35, 33].

We note that there also exist UCP results related to classes of complex coefficients, e.g. with *principally normal symbol* [31, chapter 28].

We define the class of admissible test-inclusions, used for the reconstruction methods, as

$$\mathcal{A} = \{\, C \Subset \Omega \mid C \text{ is the closure of an open set and has connected complement} \,\}.$$

Assumption 3.4.

(i) Assume there are *known* bounds on A_D, i.e. scalars $0 < \alpha < \beta$ such that

$$\alpha I \leq A_D \leq \beta I$$

in the Loewner order in Ω.

(ii) Assume that $D \Subset \Omega$ and is the closure of an open set.

(iii) Assume for every open neighborhood W of $x \in \partial D^\bullet$, there exists a relatively open connected set $V \subset D^\bullet \cap W$ that intersects $\partial D^\bullet$, satisfying either of two options:

  (a) $A_D - A_0$ is positive semidefinite in V, and there exists an open ball $B \subset V$ on which $A_D - A_0$ is uniformly positive definite.

  (b) $A_D - A_0$ is negative semidefinite in V, and there exists an open ball $B \subset V$ on which $A_D - A_0$ is uniformly negative definite.

(iv) Assume that A_0 satisfies the UCP.

We give a few remarks on these assumptions.

Remark 3.5 (Remarks on Assumption 3.4).

(a) In case $\Gamma = \partial\Omega$, then $C \Subset \Omega$ in the definition of $\mathcal{A}$ can be replaced by $C \subseteq \overline{\Omega}$, and $D \Subset \Omega$ in Assumption 3.4(ii) can be replaced by $D \subseteq \overline{\Omega}$. Also, here "connected complement" is in $\mathbb{R}^d$.

(b) Note that Assumption 3.4(iii) only relates to what happens close to $\partial D^\bullet$, so further into D we may have that $A_D - A_0$ is e.g. indefinite. Moreover, the definiteness condition is not exactly at $\partial D^\bullet$ but just close to $\partial D^\bullet$, meaning that there can be a continuous deviation from A_0 rather than a jump condition; in the isotropic case this mimics the condition of a strict local increase/decrease from the background conductivity when entering D from the outside [15].

For a measurable set C, we define the test-coefficients

$$A_C^- = \begin{cases} \alpha I & \text{in } C \\ A_0 & \text{in } \Omega \setminus C \end{cases} \quad \text{and} \quad A_C^+ = \begin{cases} \beta I & \text{in } C \\ A_0 & \text{in } \Omega \setminus C, \end{cases}$$

and the corresponding test-operators

$$\Lambda_C^- = \Lambda(A_C^-) \quad \text{and} \quad \Lambda_C^+ = \Lambda(A_C^+).$$

Theorem 3.6. *Under Assumption 3.4(i), for any measurable $C \subseteq \overline{\Omega}$ we have*

$$D \subseteq C \quad \textit{implies} \quad \Lambda_C^- \geq \Lambda(A_D) \geq \Lambda_C^+.$$

Under all of Assumption 3.4, for any $C \in \mathcal{A}$ we have

$$\Lambda_C^- \geq \Lambda(A_D) \geq \Lambda_C^+ \quad \textit{implies} \quad D \subseteq C.$$

Proof. The proof is given in Section 7. □

We also introduce linearized versions of the test-operators, which are more suitable for numerical computations. Note that for $A \in \mathcal{H}_{\mathrm{SA}}(\Omega)$ and $B \in L^\infty(\Omega)^{d\times d}$, the Frechét derivative of Λ evaluated at A and in direction B is given by (c.f. [14]):

$$\langle f, \mathrm{D}\Lambda(A;B)f\rangle = -\int_\Omega B^*\nabla u_f^A \cdot \overline{\nabla u_f^A}\,\mathrm{d}x.$$

To shorten the notation in the following result, we introduce the test-operators

$$\mathrm{D}\Lambda_C^+ = \mathrm{D}\Lambda\big(A_0; (\beta I - A_0)\chi_C\big),$$
$$\mathrm{D}\Lambda_C^- = \mathrm{D}\Lambda\big(A_0; (A_0 - \tfrac{\beta^2}{\alpha} I)\chi_C\big),$$

where χ_C is a characteristic function on the set C.

Theorem 3.7. *Pick $0 < \alpha < \beta$ such that Assumption 3.4(i) holds for both A_D and A_0. For any measurable $C \subseteq \overline{\Omega}$ we have*

$$D \subseteq C \quad \textit{implies} \quad \mathrm{D}\Lambda_C^- \geq \Lambda(A_D) - \Lambda(A_0) \geq \mathrm{D}\Lambda_C^+.$$

Under all of Assumption 3.4, for any $C \in \mathcal{A}$ we have

$$\mathrm{D}\Lambda_C^- \geq \Lambda(A_D) - \Lambda(A_0) \geq \mathrm{D}\Lambda_C^+ \quad \textit{implies} \quad D \subseteq C.$$

Proof. The proof is given in Section 8. □

Remark 3.8. Under the given assumptions, Theorem 3.6 gives

$$D^\bullet = \cap\{\, C \in \mathcal{A} \mid \Lambda_C^- \geq \Lambda(A_D) \geq \Lambda_C^+ \,\},$$

while Theorem 3.7 gives

$$D^\bullet = \cap\{\, C \in \mathcal{A} \mid \mathrm{D}\Lambda_C^- \geq \Lambda(A_D) - \Lambda(A_0) \geq \mathrm{D}\Lambda_C^+ \,\}.$$

Moreover, from the proofs we also conclude:

- If in Assumption 3.4(iii) there is only positive definiteness near $\partial D^\bullet$, we only need to check the inequalities $\Lambda(A_D) \geq \Lambda_C^+$ or $\Lambda(A_D) - \Lambda(A_0) \geq \mathrm{D}\Lambda_C^+$.
- If in Assumption 3.4(iii) there is only negative definiteness near $\partial D^\bullet$, we only need to check the inequalities $\Lambda_C^- \geq \Lambda(A_D)$ or $\mathrm{D}\Lambda_C^- \geq \Lambda(A_D) - \Lambda(A_0)$.

4. Inner reconstruction of uniformly definite anisotropic inclusions

Under much stronger definiteness assumptions (not just near $\partial D^\bullet$ as before), we can also use the inner approach of the monotonicity method. We focus on the linearized version, so that the UCP requirement is only related to the background conductivity A_0.

Assumption 4.1.

(i) Assume that we have either:
  (a) *Positive definite inclusions*, $A_D - A_0 \geq cI$ in the Loewner order in D, for known $c > 0$.
  (b) *Negative definite inclusions*, $A_D - A_0 \leq -cI$ in the Loewner order in D, for known $c > 0$.

(ii) Assume that $D \Subset \Omega$ and is the closure of an open set.

(iii) Assume that A_0 satisfies the UCP.

Again, to shorten the notation in the following results, we introduce the test-operators

$$\mathrm{D}\widehat{\Lambda}_B^+ = \mathrm{D}\Lambda\big(A_0; c(\tfrac{\alpha}{\beta})^2\chi_B I\big),$$
$$\mathrm{D}\widehat{\Lambda}_B^- = \mathrm{D}\Lambda\big(A_0; -c\chi_B I\big),$$

where χ_B is a characteristic function on the set B; note the differences compared to the previous section.

Theorem 4.2. *Assume that A_D has* positive definite inclusions *in D, corresponding to case (a) of Assumption 4.1(i). Let $A_0 \geq \alpha I$ and assume $A_D \leq \beta I$ in the Loewner order in Ω, for $\alpha, \beta > 0$. For any measurable $B \subseteq \Omega$ we have*

$$B \subseteq D \quad \textit{implies} \quad \Lambda(A_D) - \Lambda(A_0) \leq \mathrm{D}\widehat{\Lambda}_B^+.$$

Under all of Assumption 4.1, for any open set $B \subseteq \Omega$ we have

$$\Lambda(A_D) - \Lambda(A_0) \leq \mathrm{D}\widehat{\Lambda}_B^+ \quad \textit{implies} \quad B \subseteq D^\bullet.$$

Proof. The proof is given in Section 9. □

Theorem 4.3. *Assume that A_D has* negative definite inclusions *in D, corresponding to case (b) of Assumption 4.1(i). For any measurable $B \subseteq \Omega$ we have*

$$B \subseteq D \quad \textit{implies} \quad \mathrm{D}\widehat{\Lambda}_B^- \leq \Lambda(A_D) - \Lambda(A_0).$$

Under all of Assumption 4.1, for any open set $B \subseteq \Omega$ we have

$$\mathrm{D}\widehat{\Lambda}_B^- \leq \Lambda(A_D) - \Lambda(A_0) \quad \textit{implies} \quad B \subseteq D^\bullet.$$

Proof. The proof is given in Section 10. □

Remark 4.4. Under the given assumptions, for positive definite inclusions, Theorem 4.2 gives

$$D^\circ \subseteq \cup\{\, B \subset \Omega \text{ open ball} \mid \Lambda(A_D) - \Lambda(A_0) \leq \mathrm{D}\widehat{\Lambda}_B^+ \,\} \subset D^\bullet,$$

while for negative definite inclusions, Theorem 4.3 gives

$$D^\circ \subseteq \cup\{\, B \subset \Omega \text{ open ball} \mid \mathrm{D}\widehat{\Lambda}_B^- \leq \Lambda(A_D) - \Lambda(A_0) \,\} \subset D^\bullet.$$

One can of course replace "open ball" by some other basis of open sets for the Euclidean topology.

5. Operator inequalities for Neumann-to-Dirichlet maps

Although we will only consider coefficients from $\mathcal{H}_{\mathrm{SA}}(\Omega)$ in the reconstruction methods, we still prove the following more general inequalities based on $\mathcal{H}(\Omega)$ for possible future reference. See also [24, Lemma 2.1] for the isotropic case with complex conductivity.

Theorem 5.1. *Let $A_1, A_2 \in \mathcal{H}(\Omega)$ and $f \in L^2_\diamond(\Gamma)$. For $j \in \{1,2\}$ we denote $\Lambda_j = \Lambda(A_j)$ and $u_j = u_f^{A_j}$. Assume $\kappa \in \mathbb{C}$ is such that $\kappa A_1 \in \mathcal{H}(\Omega)$. We also define*

$$\mathcal{B}_\kappa = (\kappa A_1)^{\mathrm{I}}[(\kappa A_1)^{\mathrm{R}}]^{-1}(\kappa A_1)^{\mathrm{I}},$$
$$\mathcal{C}_\kappa = (\kappa A_2)^{\mathrm{I}}[(\kappa A_1)^{\mathrm{R}}]^{-1}(\kappa A_2)^{\mathrm{I}} + \mathrm{i}\big[(\kappa A_2)^{\mathrm{R}}[(\kappa A_1)^{\mathrm{R}}]^{-1}(\kappa A_2)^{\mathrm{I}} - (\kappa A_2)^{\mathrm{I}}[(\kappa A_1)^{\mathrm{R}}]^{-1}(\kappa A_2)^{\mathrm{R}}\big].$$

Then

$$\langle f, [\overline{\kappa}(\Lambda_1-\Lambda_2)]^{\mathrm{R}} f\rangle \geq \int_\Omega \big[[\kappa(A_2-A_1)]^{\mathrm{R}} - \mathcal{B}_\kappa\big]\nabla u_2\cdot\overline{\nabla u_2}\,\mathrm{d}x, \tag{5.1}$$

$$\langle f, [\overline{\kappa}(\Lambda_1-\Lambda_2)]^{\mathrm{R}} f\rangle \leq \int_\Omega \big[(\kappa A_2)^{\mathrm{R}}[(\kappa A_1)^{\mathrm{R}}]^{-1}[\kappa(A_2-A_1)]^{\mathrm{R}} + \mathcal{C}_\kappa\big]\nabla u_2\cdot\overline{\nabla u_2}\,\mathrm{d}x. \tag{5.2}$$

Proof. We start by proving (5.1). A computation reveals that

$$(\kappa A_1)[(\kappa A_1)^{\mathrm{R}}]^{-1}(\kappa A_1)^* = (\kappa A_1)^{\mathrm{R}} + \mathcal{B}_\kappa,$$

which is used below, combined with (2.3) and (2.4):

$$\begin{aligned}
0 &\leq \int_\Omega \big|[(\kappa A_1)^{\mathrm{R}}]^{1/2}\nabla u_1 - [(\kappa A_1)^{\mathrm{R}}]^{-1/2}(\kappa A_1)^*\nabla u_2\big|^2\,\mathrm{d}x \\
&= \mathrm{Re}\Big(\kappa\int_\Omega A_1\nabla u_1\cdot\overline{\nabla u_1}\,\mathrm{d}x\Big) - 2\,\mathrm{Re}\Big(\kappa\int_\Omega A_1\nabla u_1\cdot\overline{\nabla u_2}\,\mathrm{d}x\Big) \\
&\quad + \int_\Omega\big[(\kappa A_1)^{\mathrm{R}} + \mathcal{B}_\kappa\big]\nabla u_2\cdot\overline{\nabla u_2}\,\mathrm{d}x \\
&= \langle f, [\overline{\kappa}(\Lambda_1-\Lambda_2)]^{\mathrm{R}} f\rangle + \int_\Omega\big[(\kappa A_1)^{\mathrm{R}} - (\kappa A_2)^{\mathrm{R}} + \mathcal{B}_\kappa\big]\nabla u_2\cdot\overline{\nabla u_2}\,\mathrm{d}x.
\end{aligned}$$

Next we prove (5.2). A computation reveals that

$$\begin{aligned}
(\kappa A_2)^*[(\kappa A_1)^{\mathrm{R}}]^{-1}(\kappa A_2) &= (\kappa A_2)^{\mathrm{R}}[(\kappa A_1)^{\mathrm{R}}]^{-1}(\kappa A_2)^{\mathrm{R}} + \mathcal{C}_\kappa \\
&= (\kappa A_2)^{\mathrm{R}} + (\kappa A_2)^{\mathrm{R}}[(\kappa A_1)^{\mathrm{R}}]^{-1}[\kappa(A_2-A_1)]^{\mathrm{R}} + \mathcal{C}_\kappa,
\end{aligned}$$

which is used below, combined with (2.3) and (2.4):

$$\begin{aligned}
0 &\leq \int_\Omega \big|[(\kappa A_1)^{\mathrm{R}}]^{1/2}\nabla u_1 - [(\kappa A_1)^{\mathrm{R}}]^{-1/2}(\kappa A_2)\nabla u_2\big|^2\,\mathrm{d}x \\
&= \mathrm{Re}\Big(\kappa\int_\Omega A_1\nabla u_1\cdot\overline{\nabla u_1}\,\mathrm{d}x\Big) - 2\,\mathrm{Re}\Big(\kappa\int_\Omega A_2\nabla u_2\cdot\overline{\nabla u_1}\,\mathrm{d}x\Big) \\
&\quad + \int_\Omega\big[(\kappa A_2)^{\mathrm{R}} + (\kappa A_2)^{\mathrm{R}}[(\kappa A_1)^{\mathrm{R}}]^{-1}[\kappa(A_2-A_1)]^{\mathrm{R}} + \mathcal{C}_\kappa\big]\nabla u_2\cdot\overline{\nabla u_2}\,\mathrm{d}x \\
&= \langle f, [\overline{\kappa}(\Lambda_2-\Lambda_1)]^{\mathrm{R}} f\rangle + \int_\Omega\big[(\kappa A_2)^{\mathrm{R}}[(\kappa A_1)^{\mathrm{R}}]^{-1}[\kappa(A_2-A_1)]^{\mathrm{R}} + \mathcal{C}_\kappa\big]\nabla u_2\cdot\overline{\nabla u_2}\,\mathrm{d}x. \qquad \square
\end{aligned}$$

Remark 5.2. Note that in the isotropic case, the large square bracket in $\mathcal{C}_\kappa$ vanishes (after the imaginary unit). Also, if $\kappa A_1 \in \mathcal{H}_{\mathrm{SA}}(\Omega)$ then $\mathcal{B}_\kappa = 0$ and if $\kappa A_2 \in \mathcal{H}_{\mathrm{SA}}(\Omega)$ then $\mathcal{C}_\kappa = 0$.

We get some corollaries of Theorem 5.1, based on the choice of κ.

Corollary 5.3. *In the setting of Theorem 5.1 with $\kappa = 1$, we have*

$$\begin{aligned}
\langle f, (\Lambda_1^{\mathrm{R}} - \Lambda_2^{\mathrm{R}})f\rangle &\geq \int_\Omega\big[A_2^{\mathrm{R}} - A_1^{\mathrm{R}} - \mathcal{B}_1\big]\nabla u_2\cdot\overline{\nabla u_2}\,\mathrm{d}x, \\
\langle f, (\Lambda_1^{\mathrm{R}} - \Lambda_2^{\mathrm{R}})f\rangle &\leq \int_\Omega\big[A_2^{\mathrm{R}}(A_1^{\mathrm{R}})^{-1}(A_2^{\mathrm{R}} - A_1^{\mathrm{R}}) + \mathcal{C}_1\big]\nabla u_2\cdot\overline{\nabla u_2}\,\mathrm{d}x.
\end{aligned}$$

Corollary 5.4. *In the setting of Theorem 5.1 with $\kappa = -\mathrm{i}$, provided that $A_1^{\mathrm{I}} \in \mathcal{H}_{\mathrm{SA}}(\Omega)$, we have*

$$\begin{aligned}
\langle f, (\Lambda_2^{\mathrm{I}} - \Lambda_1^{\mathrm{I}})f\rangle &\geq \int_\Omega\big[A_2^{\mathrm{I}} - A_1^{\mathrm{I}} - \mathcal{B}_{-\mathrm{i}}\big]\nabla u_2\cdot\overline{\nabla u_2}\,\mathrm{d}x, \\
\langle f, (\Lambda_2^{\mathrm{I}} - \Lambda_1^{\mathrm{I}})f\rangle &\leq \int_\Omega\big[A_2^{\mathrm{I}}(A_1^{\mathrm{I}})^{-1}(A_2^{\mathrm{I}} - A_1^{\mathrm{I}}) + \mathcal{C}_{-\mathrm{i}}\big]\nabla u_2\cdot\overline{\nabla u_2}\,\mathrm{d}x.
\end{aligned}$$

What we will need for the monotonicity method is the following, corresponding to the special case of Theorem 5.1 where $A_1, A_2 \in \mathcal{H}_{\mathrm{SA}}(\Omega)$ and $\kappa = 1$. We also provide a slightly different proof that is arguably easier to read.

Proposition 5.5. *Let $A_1, A_2 \in \mathcal{H}_{\mathrm{SA}}(\Omega)$ and $f \in L^2_\diamond(\Gamma)$. For $j \in \{1,2\}$ we denote $\Lambda_j = \Lambda(A_j)$ and $u_j = u_f^{A_j}$. Then*

$$\langle f, (\Lambda_1 - \Lambda_2) f\rangle \geq \int_\Omega (A_2 - A_1)\nabla u_2 \cdot \overline{\nabla u_2}\,\mathrm{d}x, \tag{5.3}$$

$$\langle f, (\Lambda_1 - \Lambda_2) f\rangle \leq \int_\Omega A_2 A_1^{-1}(A_2 - A_1)\nabla u_2 \cdot \overline{\nabla u_2}\,\mathrm{d}x. \tag{5.4}$$

Proof. We start by proving (5.3), which is by the computation below combined with (2.3),

$$\begin{aligned}
&\int_\Omega (A_2 - A_1)\nabla u_2 \cdot \overline{\nabla u_2}\,\mathrm{d}x \\
&\qquad \leq \int_\Omega (A_2 - A_1)\nabla u_2 \cdot \overline{\nabla u_2}\,\mathrm{d}x + \int_\Omega |A_1^{1/2}\nabla(u_1 - u_2)|^2\,\mathrm{d}x \\
&\qquad = \int_\Omega A_2\nabla u_2 \cdot \overline{\nabla u_2}\,\mathrm{d}x + \int_\Omega A_1\nabla u_1 \cdot \overline{\nabla u_1}\,\mathrm{d}x - 2\,\mathrm{Re}\Big(\int_\Omega A_1\nabla u_1 \cdot \overline{\nabla u_2}\,\mathrm{d}x\Big) \\
&\qquad = \langle f, (\Lambda_1 - \Lambda_2) f\rangle.
\end{aligned}$$

Next we prove (5.4). Using that

$$A_2A_1^{-1}(A_2 - A_1) = A_2A_1^{-1}A_2 - A_2$$

is self-adjoint, we get

$$\begin{aligned}
&\int_\Omega A_2A_1^{-1}(A_2 - A_1)\nabla u_2 \cdot \overline{\nabla u_2}\,\mathrm{d}x \\
&\qquad \geq \int_\Omega A_2A_1^{-1}(A_2 - A_1)\nabla u_2 \cdot \overline{\nabla u_2}\,\mathrm{d}x - \int_\Omega |A_1^{1/2}\nabla u_1 - A_1^{-1/2}A_2\nabla u_2|^2\,\mathrm{d}x \\
&\qquad = -\int_\Omega A_2\nabla u_2 \cdot \overline{\nabla u_2}\,\mathrm{d}x - \int_\Omega A_1\nabla u_1 \cdot \overline{\nabla u_1}\,\mathrm{d}x + 2\,\mathrm{Re}\Big(\int_\Omega A_2\nabla u_2 \cdot \overline{\nabla u_1}\,\mathrm{d}x\Big) \\
&\qquad = \langle f, (\Lambda_1 - \Lambda_2) f\rangle.
\end{aligned}$$

□

Note that the first inequality in Proposition 5.5 implies monotonicity of the ND maps, since

$$\int_\Omega (A_2 - A_1)\nabla u_2 \cdot \overline{\nabla u_2}\,\mathrm{d}x \leq \langle f, (\Lambda_1 - \Lambda_2) f\rangle \leq \int_\Omega (A_2 - A_1)\nabla u_1 \cdot \overline{\nabla u_1}\,\mathrm{d}x. \tag{5.5}$$

It is not immediately clear how one would make use of the second inequality in Proposition 5.5, due to the matrix product in the front. What we need to consider is when $A_2 - A_1$ is positive/negative (semi)definite, if we can also infer that $A_2A_1^{-1}(A_2 - A_1)$ has the same properties. This is the result of the following proposition.

Proposition 5.6. *Let $c \geq 0$ and $V \subseteq \Omega$. Let $A_1, A_2 \in \mathcal{H}_{\mathrm{SA}}(\Omega)$ with $A_1 \leq \beta I$ and $A_2 \geq \alpha I$ in V for $\alpha, \beta > 0$.*

(i) *If $A_2 - A_1 \geq cI$ in V, then $A_2A_1^{-1}(A_2 - A_1) \geq cI$ in V.*
(ii) *If $A_2 - A_1 \leq -cI$ in V, then $A_2A_1^{-1}(A_2 - A_1) \leq -c(\frac{\alpha}{\beta})^2 I$ in V.*

Proof. Let $A_2 - A_1 \geq cI$ in the Loewner order in V for some $c \geq 0$. Let $\xi \in \mathbb{C}^d$ and note that A_1^{-1} is positive definite. Now we put the pieces together:

$$\begin{aligned}
A_2A_1^{-1}(A_2 - A_1)\xi \cdot \overline{\xi} &= (A_2 - A_1)A_1^{-1}(A_2 - A_1)\xi \cdot \overline{\xi} + (A_2 - A_1)\xi \cdot \overline{\xi} \\
&= A_1^{-1}(A_2 - A_1)\xi \cdot \overline{(A_2 - A_1)\xi} + (A_2 - A_1)\xi \cdot \overline{\xi} \\
&\geq c|\xi|^2,
\end{aligned}$$

which concludes the proof of (i).

For (ii), we first prove the assertion that if

$$A_1 - A_2 \geq cI \tag{5.6}$$

for some $c \geq 0$, then

$$A_2^{-1} - A_1^{-1} \geq c\beta^{-2} I. \tag{5.7}$$

The semidefinite case ($c = 0$) of (5.7) is a simple consequence of $t \mapsto -t^{-1}$ being an operator monotone function, c.f. [8, chapter 7]. To prove (5.7), first note that we have

$$\begin{aligned} A_2^{-1} - A_1^{-1} &= A_1^{-1}(A_1 - A_2)A_2^{-1} \\ &= A_1^{-1}(A_1 - A_2)A_1^{-1} + A_1^{-1}(A_1 - A_2)(A_2^{-1} - A_1^{-1}) \\ &= A_1^{-1}\big[A_1 - A_2 + (A_1 - A_2)A_2^{-1}(A_1 - A_2)\big]A_1^{-1}. \end{aligned}$$

Note that $A_1^{-1} \geq \beta^{-1}I$ and likewise that A_2^{-1} is positive definite. Hence, we have

$$\begin{aligned} (A_2^{-1} - A_1^{-1})\xi \cdot \overline{\xi} &= (A_1 - A_2)A_1^{-1}\xi \cdot \overline{A_1^{-1}\xi} + A_2^{-1}(A_1 - A_2)A_1^{-1}\xi \cdot \overline{(A_1 - A_2)A_1^{-1}\xi} \\ &\geq c|A_1^{-1}\xi|^2 \geq c\beta^{-2}|\xi|^2. \end{aligned}$$

Now assume that in V we have (5.6) for some $c \geq 0$. Then (5.7) implies

$$A_2A_1^{-1}(A_1 - A_2)\xi \cdot \overline{\xi} = (A_2^{-1} - A_1^{-1})A_2\xi \cdot \overline{A_2\xi} \geq c\beta^{-2}|A_2\xi|^2 \geq c(\tfrac{\alpha}{\beta})^2|\xi|^2,$$

which concludes the proof of (ii). □

6. Localized potentials for anisotropic coefficients

We extend localization results from [20] to anisotropic coefficients in $\mathcal{H}(\Omega)$. For a measurable set $V \subseteq \overline{\Omega}$ and $F \in L^2(V)^d$, we define $w = w_F^A$ as the unique solution in $H^1_\diamond(\Omega)$ to the variational problem

$$\int_\Omega A^*\nabla w \cdot \overline{\nabla v} \, \mathrm{d}x = \int_V F \cdot \overline{\nabla v} \tag{6.1}$$

for all $v \in H^1_\diamond(\Omega)$. We define the *virtual measurement operator* $L_V(A)\colon L^2(V)^d \to L^2_\diamond(\Gamma)$ as

$$L_V(A)F = w_F^A|_\Gamma.$$

In the following, R is used to denote the range of an operator.

Proposition 6.1. *For $A \in \mathcal{H}(\Omega)$ and measurable sets $V, V_1, V_2 \subseteq \overline{\Omega}$, there are the following properties of the virtual measurement operator:*

(i) *$R(L_V(A))$ is independent of the function values $A|_V$,*
(ii) *$V_1 \subseteq V_2$ implies $R(L_{V_1}(A)) \subseteq R(L_{V_2}(A))$,*
(iii) *The adjoint $L_V(A)^*\colon L^2_\diamond(\Gamma) \to L^2(V)^d$ is given by $L_V(A)^*f = \nabla u_f^A|_V$.*

Proof.

(i): Let $A_1, A_2 \in \mathcal{H}(\Omega)$ such that $A_1 = A_2$ outside V. We will prove that $R(L_V(A_1)) = R(L_V(A_2))$. Let $\varphi \in R(L_V(A_1))$, i.e. $\varphi = w|_\Gamma$ with $w = w_{F_1}^{A_1}$ for some $F_1 \in L^2(V)^d$. Now define $F_2 = (A_2 - A_1)^*\nabla w|_V \in L^2(V)^d$. Then (6.1) gives

$$\begin{aligned} \int_\Omega A_2^*\nabla w \cdot \overline{\nabla v} \, \mathrm{d}x &= \int_\Omega (A_2 - A_1)^*\nabla w \cdot \overline{\nabla v} \, \mathrm{d}x + \int_\Omega A_1^*\nabla w \cdot \overline{\nabla v} \, \mathrm{d}x \\ &= \int_V (A_2 - A_1)^*\nabla w \cdot \overline{\nabla v} \, \mathrm{d}x + \int_V F_1 \cdot \overline{\nabla v} \, \mathrm{d}x \\ &= \int_V (F_1 + F_2) \cdot \overline{\nabla v} \, \mathrm{d}x \end{aligned}$$

for all $v \in H^1_\diamond(\Omega)$. Hence, we have that $\varphi = L_V(A_2)(F_1+F_2)$ so indeed $\varphi \in R(L_V(A_2))$. Repeating the above proof with A_1 and A_2 interchanged results in $R(L_V(A_1)) = R(L_V(A_2))$ as desired.

(ii): Assume $V_1 \subseteq V_2$, and let $\varphi = L_{V_1}(A)F = w|_\Gamma$ for some $F \in L^2(V_1)^d$. Let $\widetilde{F}$ denote the extension by zero of F on the set V_2. Then

$$\int_\Omega A^*\nabla w \cdot \overline{\nabla v} \, \mathrm{d}x = \int_{V_1} F \cdot \overline{\nabla v} \, \mathrm{d}x = \int_{V_2} \widetilde{F} \cdot \overline{\nabla v} \, \mathrm{d}x$$

for all $v \in H^1_\diamond(\Omega)$. Hence, we also have $\varphi = L_{V_2}(A)\widetilde{F}$, i.e. $R(L_{V_1}(A)) \subseteq R(L_{V_2}(A))$.

(iii): Let $F \in L^2(V)^d$ and $f \in L^2_\diamond(\Gamma)$, then from the variational forms (6.1) and (2.2), for $w = w_F^A$ and $u = u_f^A$, we obtain

$$\begin{aligned}\langle L_V(A)^* f, F\rangle_{L^2(V)^d} &= \langle f, L_V(A)F\rangle_{L^2(\Gamma)} = \langle f, w\rangle_{L^2(\Gamma)}\\ &= \int_\Omega A\nabla u\cdot\overline{\nabla w}\,\mathrm{d}x = \int_V \overline{F}\cdot\nabla u\,\mathrm{d}x = \langle \nabla u|_V, F\rangle_{L^2(V)^d}.\end{aligned}$$

Since this holds for all $F \in L^2(V)^d$, we conclude that $L_V(A)^* f = \nabla u|_V$. □

We note that range inclusions imply norm inequalities for the adjoint operators.

Lemma 6.2 (Lemma 2.5 in [20])**.** *Let H, K_1, and K_2 be Hilbert spaces and let $T_j \in \mathscr{L}(K_j, H)$ for $j \in \{1,2\}$. Then*

$$R(T_1) \subseteq R(T_2) \qquad \textit{if and only if} \qquad \exists C>0,\, \forall x\in H\colon \|T_1^* x\|_{K_1} \le C\|T_2^* x\|_{K_2}.$$

Finally, we are able to prove the existence of localized potentials.

Theorem 6.3. *Let $U \subset \overline{\Omega}$ be a relatively open connected set that intersects Γ and let $B \subset U$ be a non-empty open set. Assume $A \in \mathcal{H}(\Omega)$ where both A and A^* satisfy the UCP in U. Then there are sequences (f_j) in $L^2_\diamond(\Gamma)$ and (u_j) in $H^1_\diamond(\Omega)$, with $u_j = u^A_{f_j}$, such that*

$$\lim_{j\to\infty}\int_B |\nabla u_j|^2\,\mathrm{d}x = \infty \qquad \textit{and} \qquad \lim_{j\to\infty}\int_{\Omega\setminus U}|\nabla u_j|^2\,\mathrm{d}x = 0.$$

Proof. We let $V_1 \subseteq B \subset V \subseteq U$ where V_1 and V are relatively open non-empty connected sets, and define $V_2 = \Omega\setminus\overline{V}$. These sets are chosen such that V intersects Γ, $\mathrm{dist}(V_1, V_2) > 0$, and that $\partial(\Omega\setminus\overline{V_j})$ is Lipschitz continuous.

We start by proving that $R(L_{V_1}(A))$ and $R(L_{V_2}(A))$ have a trivial intersection. Hence, let $\varphi \in R(L_{V_1}(A)) \cap R(L_{V_2}(A))$, i.e. $\varphi = w_j|_\Gamma$ with $w_j = w^A_{F_j}$ for some $F_j \in L^2(V_j)^d$ for $j\in\{1,2\}$. From (6.1) we conclude that

$$\int_\Omega A^*\nabla w_j\cdot\overline{\nabla v}\,\mathrm{d}x = 0 \text{ for all } v \in H^1(\Omega\setminus\overline{V_j}).$$

From the weak form of the conductivity problem (2.2), with conductivity A^*, we thus have

$$\begin{aligned}-\nabla\cdot(A^*\nabla w_j) &= 0 \text{ in } \Omega\setminus\overline{V_j}\\ \nu\cdot(A^*\nabla w_j) &= 0 \text{ on } \partial(\Omega\setminus\overline{V_j}),\end{aligned}$$

in particular $\nu\cdot(A^*\nabla w_j) \equiv 0$ on $\partial V\cap\Gamma$, i.e. the Cauchy data of w_1 and w_2 coincide on $\partial V\cap\Gamma$. Unique continuation entails that $w_1 = w_2$ in $V\setminus\overline{V_1}$. We may now "glue" w_1 and w_2 together, using that they agree in $V\setminus\overline{V_1}$:

$$u = \begin{cases} w_1 & \text{in } \Omega\setminus\overline{V_1},\\ w_2 & \text{in } V_1.\end{cases}$$

Now we have $-\nabla\cdot(A^*\nabla u) = 0$ in Ω and $\nu\cdot(A^*\nabla u) = \nu\cdot(A^*\nabla w_1) \equiv 0$ on $\partial\Omega$. By uniqueness of such a solution in $H^1_\diamond(\Omega)$, it follows that $u\equiv 0$ in Ω, so in particular $\varphi = w_1|_\Gamma = u|_\Gamma \equiv 0$, i.e.

$$R(L_{V_1}(A)) \cap R(L_{V_2}(A)) = \{0\}.$$

Next we prove that $R(L_{V_1}(A))$ is non-trivial. Suppose that $0 \equiv L_{V_1}(A)^* f = \nabla u^A_f|_{V_1}$, then u^A_f is a constant function in V_1, and also in V by unique continuation. In particular, $f = \nu\cdot(A\nabla u^A_f)|_\Gamma \equiv 0$ on $\partial V\cap\Gamma$. Thus, $L_{V_1}(A)^*$ is injective on

$$W = \{\, f \in L^2_\diamond(\Gamma) \mid \operatorname{supp} f \subseteq \partial V\cap\Gamma \,\}.$$

Since $\overline{R(L_{V_1}(A))} = N(L_{V_1}(A)^*)^\perp$ then $R(L_{V_1}(A))$ contains a dense subset of W and must be non-trivial.

In total we have that $R(L_{V_1}(A)) \not\subseteq R(L_{V_2}(A))$. From Lemma 6.2 this implies the existence of a sequence (f_j) in $L^2_\diamond(\Gamma)$ such that $u_j = u^A_{f_j}$ satisfy (c.f. Proposition 6.1(iii)):

$$\int_B |\nabla u_j|^2\,\mathrm{d}x \ge \int_{V_1}|\nabla u_j|^2\,\mathrm{d}x = \|L_{V_1}(A)^* f_j\|^2_{L^2(V_1)^d} \to \infty$$

and

$$\int_{\Omega\setminus U} |\nabla u_j|^2 \,\mathrm{d}x \leq \int_{V_2} |\nabla u_j|^2 \,\mathrm{d}x = \|L_{V_2}(A)^* f_j\|^2_{L^2(V_2)^d} \to 0$$

for $j \to \infty$. □

The following result shows that the sequence (f_j) of Neumann conditions (e.g. from Theorem 6.3) such that the corresponding electric power $|\nabla u_j|^2$ is localized, can be chosen independently of the function values of A on the set where the power tends to zero.

Proposition 6.4. *Let B and U be as in Theorem 6.3, and assume for $A_1 \in \mathcal{H}(\Omega)$ and (f_j) in $L^2_\diamond(\Gamma)$ that $u_j = u^{A_1}_{f_j}$ satisfy*

$$\lim_{j\to\infty} \int_B |\nabla u_j|^2 \,\mathrm{d}x = \infty \qquad \text{and} \qquad \lim_{j\to\infty} \int_{\Omega\setminus U} |\nabla u_j|^2 \,\mathrm{d}x = 0.$$

If $A_2 \in \mathcal{H}(\Omega)$ with $\operatorname{supp}(A_1 - A_2) \subseteq \overline{\Omega} \setminus U$, then $\widehat{u}_j = u^{A_2}_{f_j}$ also satisfy

$$\lim_{j\to\infty} \int_B |\nabla \widehat{u}_j|^2 \,\mathrm{d}x = \infty \qquad \text{and} \qquad \lim_{j\to\infty} \int_{\Omega\setminus U} |\nabla \widehat{u}_j|^2 \,\mathrm{d}x = 0.$$

Proof. Let $V_1 = B$ and $V_2 = \overline{\Omega} \setminus U$. From Proposition 6.1(iii) it amounts to proving

$$\lim_{j\to\infty} \|L_{V_1}(A_2)^* f_j\|_{L^2(V_1)^d} = \infty \quad \text{and} \quad \lim_{j\to\infty} \|L_{V_2}(A_2)^* f_j\|_{L^2(V_2)^d} = 0.$$

From Proposition 6.1(i) we have that $R(L_{V_2}(A_1)) = R(L_{V_2}(A_2))$. Hence a combination of Proposition 6.1(iii) and Lemma 6.2 gives

$$\|L_{V_2}(A_2)^* f_j\|_{L^2(V_2)^d} \leq C \|L_{V_2}(A_1)^* f_j\|_{L^2(V_2)^d} \to 0 \text{ for } j \to \infty.$$

By the same argumentation, we also have $R(L_{V_1\cup V_2}(A_1)) = R(L_{V_1\cup V_2}(A_2))$, which implies

$$\begin{aligned}\|L_{V_1\cup V_2}(A_2)^* f_j\|_{L^2(V_1\cup V_2)^d} &\geq c \|L_{V_1\cup V_2}(A_1)^* f_j\|_{L^2(V_1\cup V_2)^d} \\ &\geq c \|L_{V_1}(A_1)^* f_j\|_{L^2(V_1)^d} \to \infty \text{ for } j \to \infty.\end{aligned}$$

Finally we combine the two above limits, using Proposition 6.1(iii),

$$\|L_{V_1}(A_2)^* f_j\|^2_{L^2(V_1)^d} = \|L_{V_1\cup V_2}(A_2)^* f_j\|^2_{L^2(V_1\cup V_2)^d} - \|L_{V_2}(A_2)^* f_j\|^2_{L^2(V_2)^d} \to \infty \text{ for } j \to \infty,$$

thereby concluding the proof. □

7. Proof of Theorem 3.6

The proof that $D \subseteq C$ implies $\Lambda_C^- \geq \Lambda(A_D) \geq \Lambda_C^+$ follows directly from Assumption 3.4(i) and (5.5), since we have

$$A_C^- \leq A_D \leq A_C^+$$

in the Loewner order in Ω.

Now assume that $D \not\subseteq C$, i.e. that $D^\bullet \not\subseteq C$. Since both sets are closures of opens sets and have connected complements, there is a relatively open set that connects $D^\bullet \setminus C$ with Γ that we may use for localization. Specifically, there exist a relatively open connected set $U \subset \overline{\Omega} \setminus C$ that intersects Γ and an open ball $B \subset D^\bullet \cap U$. Moreover, Assumption 3.4(iii) ensures that we can pick B and U such that we arrive at one of two cases:

- Case 1: $A_D - A_0$ is positive semidefinite in U and uniformly positive definite in B.
- Case 2: $A_D - A_0$ is negative semidefinite in U and uniformly negative definite in B.

In the first case we will prove that $\Lambda(A_D) \not\geq \Lambda_C^+$, and in the second case we will prove that $\Lambda_C^- \not\geq \Lambda(A_D)$. Together the two cases prove the contrapositive formulation of the assertion

$$\Lambda_C^- \geq \Lambda(A_D) \geq \Lambda_C^+ \quad \text{implies} \quad D \subseteq C.$$

Case 1: Using Theorem 6.3 and Proposition 6.4 (as $C \subset \Omega \setminus U$), there is a sequence (f_j) in $L^2_\diamond(\Gamma)$ such that $u_j = u_{f_j}^{A_C^+}$ satisfy

$$\lim_{j\to\infty} \int_B |\nabla u_j|^2 \,\mathrm{d}x = \infty \qquad \text{and} \qquad \lim_{j\to\infty} \int_{\Omega\setminus U} |\nabla u_j|^2 \,\mathrm{d}x = 0.$$

Note that $A_C^+ = A_0$ in U where $A_D - A_0$ is positive semidefinite, and also in B where $A_D - A_0$ is uniformly positive definite. From Proposition 5.6(ii) then $A_0 A_D^{-1}(A_0 - A_D)$ is negative semidefinite in U, and there exists a scalar $\tilde{c} > 0$ such that

$$A_0 A_D^{-1}(A_0 - A_D) \leq -\tilde{c} I \text{ in } B.$$

Now Proposition 5.5 gives

$$\begin{aligned}\langle f_j, \left[\Lambda(A_D) - \Lambda_C^+\right] f_j\rangle &\leq \int_\Omega A_C^+ A_D^{-1}(A_C^+ - A_D)\nabla u_j \cdot \overline{\nabla u_j} \,\mathrm{d}x \\ &\leq \|A_C^+ A_D^{-1}(A_C^+ - A_D)\|_* \int_{\Omega\setminus U} |\nabla u_j|^2 \,\mathrm{d}x - \tilde{c}\int_B |\nabla u_j|^2 \,\mathrm{d}x.\end{aligned}$$

We conclude that

$$\lim_{j\to\infty} \langle f_j, \left[\Lambda(A_D) - \Lambda_C^+\right] f_j\rangle = -\infty,$$

in particular, $\Lambda(A_D) \not\geq \Lambda_C^+$.

Case 2: Using Theorem 6.3 and Proposition 6.4 (as $C \subset \Omega \setminus U$), there is a sequence (f_j) in $L^2_\diamond(\Gamma)$ such that $u_j = u_{f_j}^{A_C^-}$ satisfy

$$\lim_{j\to\infty} \int_B |\nabla u_j|^2 \,\mathrm{d}x = \infty \qquad \text{and} \qquad \lim_{j\to\infty} \int_{\Omega\setminus U} |\nabla u_j|^2 \,\mathrm{d}x = 0.$$

Note that $A_C^- = A_0$ in U where $A_D - A_0$ is negative semidefinite, and also in B where $A_D - A_0$ is uniformly negative definite. From Proposition 5.5 we have

$$\begin{aligned}\langle f_j, \left[\Lambda_C^- - \Lambda(A_D)\right] f_j\rangle &\leq \int_\Omega (A_D - A_C^-)\nabla u_j \cdot \overline{\nabla u_j} \,\mathrm{d}x \\ &\leq \|A_D - A_C^-\|_* \int_{\Omega\setminus U} |\nabla u_j|^2 \,\mathrm{d}x - c\int_B |\nabla u_j|^2 \,\mathrm{d}x\end{aligned}$$

for a scalar $c > 0$. We conclude that

$$\lim_{j\to\infty} \langle f_j, \left[\Lambda_C^- - \Lambda(A_D)\right] f_j\rangle = -\infty,$$

in particular, $\Lambda_C^- \not\geq \Lambda(A_D)$.

8. Proof of Theorem 3.7

Assume $D \subseteq C$ and let $u_0 = u_f^{A_0}$. From Proposition 5.5 we have

$$\begin{aligned}\langle f, \left[\Lambda(A_D) - \Lambda(A_0) - \mathrm{D}\Lambda_C^+\right] f\rangle &\geq \int_\Omega (A_0 - A_D)\nabla u_0 \cdot \overline{\nabla u_0} \,\mathrm{d}x + \int_C (\beta I - A_0)\nabla u_0 \cdot \overline{\nabla u_0} \,\mathrm{d}x \\ &= \int_D (A_0 - A_D)\nabla u_0 \cdot \overline{\nabla u_0} \,\mathrm{d}x + \int_C (\beta I - A_0)\nabla u_0 \cdot \overline{\nabla u_0} \,\mathrm{d}x \\ &\geq \int_{C\setminus D} (\beta I - A_0)\nabla u_0 \cdot \overline{\nabla u_0} \,\mathrm{d}x \geq 0.\end{aligned}$$

For the next part we need an additional bound. Let $\xi \in \mathbb{C}^d$, then we have

$$A_0 A_D^{-1} A_0 \xi \cdot \overline{\xi} = A_D^{-1} A_0 \xi \cdot \overline{A_0 \xi} \leq \alpha^{-1} |A_0 \xi|^2 \leq \tfrac{\beta^2}{\alpha} |\xi|^2. \tag{8.1}$$

Note that $A_0 \leq \beta I \leq \frac{\beta^2}{\alpha} I$. Now from Proposition 5.5 and (8.1) we get

$$\begin{aligned}
&\langle f, \left[\Lambda(A_0) - \Lambda(A_D) + \mathrm{D}\Lambda_C^-\right] f\rangle \\
&\qquad \geq \int_\Omega A_0 A_D^{-1}(A_D - A_0)\nabla u_0 \cdot \overline{\nabla u_0}\,\mathrm{d}x + \int_C \left(\tfrac{\beta^2}{\alpha} I - A_0\right)\nabla u_0 \cdot \overline{\nabla u_0}\,\mathrm{d}x \\
&\qquad = \int_D (A_0 - A_0 A_D^{-1} A_0)\nabla u_0 \cdot \overline{\nabla u_0}\,\mathrm{d}x + \int_C \left(\tfrac{\beta^2}{\alpha} I - A_0\right)\nabla u_0 \cdot \overline{\nabla u_0}\,\mathrm{d}x \\
&\qquad \geq \int_{C\setminus D} \left(\tfrac{\beta^2}{\alpha} I - A_0\right)\nabla u_0 \cdot \overline{\nabla u_0}\,\mathrm{d}x \geq 0.
\end{aligned}$$

This concludes the proof for the first part of the theorem statement.

Now assume that $D \not\subseteq C$, i.e. that $D^\bullet \not\subseteq C$. Since both sets are closures of opens sets and have connected complements, there is a relatively open set that connects $D^\bullet \setminus C$ with Γ that we may use for localization. Specifically, there exist a relatively open connected set $U \subset \overline{\Omega} \setminus C$ that intersects Γ and an open ball $B \subset D^\bullet \cap U$. Moreover, Assumption 3.4(iii) ensures that we can pick B and U such that we arrive at one of two cases:

- Case 1: $A_D - A_0$ is positive semidefinite in U and uniformly positive definite in B.
- Case 2: $A_D - A_0$ is negative semidefinite in U and uniformly negative definite in B.

In the first case we will prove that $\Lambda(A_D) - \Lambda(A_0) \not\geq \mathrm{D}\Lambda_C^+$, and in the second case we will prove that $\mathrm{D}\Lambda_C^- \not\geq \Lambda(A_D) - \Lambda(A_0)$. Together the two cases prove the contrapositive formulation of the assertion

$$\mathrm{D}\Lambda_C^- \geq \Lambda(A_D) - \Lambda(A_0) \geq \mathrm{D}\Lambda_C^+ \quad \text{implies} \quad D \subseteq C.$$

For both cases we use the same sequence of localized potentials. From Theorem 6.3 there is a sequence (f_j) in $L^2_\diamond(\Gamma)$ such that $u_j = u_{f_j}^{A_0}$ satisfy

$$\lim_{j\to\infty} \int_B |\nabla u_j|^2\,\mathrm{d}x = \infty \qquad \text{and} \qquad \lim_{j\to\infty} \int_{\Omega\setminus U} |\nabla u_j|^2\,\mathrm{d}x = 0.$$

Case 1: We now combine Proposition 5.5 with Proposition 5.6(ii), using the facts that $A_D - A_0$ is positive semidefinite in U and $A_D - A_0 \geq cI$ in B for some scalar $c > 0$, which results in

$$\begin{aligned}
&\langle f_j, \left[\Lambda(A_D) - \Lambda(A_0) - \mathrm{D}\Lambda_C^+\right] f_j\rangle \\
&\quad \leq \int_\Omega A_0 A_D^{-1}(A_0 - A_D)\nabla u_j \cdot \overline{\nabla u_j}\,\mathrm{d}x + \int_C (\beta I - A_0)\nabla u_j \cdot \overline{\nabla u_j}\,\mathrm{d}x \\
&\quad \leq \|A_0 A_D^{-1}(A_0 - A_D)\|_* \int_{\Omega\setminus U} |\nabla u_j|^2\,\mathrm{d}x - c\left(\tfrac{\alpha}{\beta}\right)^2 \int_B |\nabla u_j|^2\,\mathrm{d}x + \|\beta I - A_0\|_* \int_C |\nabla u_j|^2\,\mathrm{d}x.
\end{aligned}$$

Since $C \subset \Omega \setminus U$, we conclude that

$$\lim_{j\to\infty} \langle f_j, \left[\Lambda(A_D) - \Lambda(A_0) - \mathrm{D}\Lambda_C^+\right] f_j\rangle = -\infty,$$

in particular, $\Lambda(A_D) - \Lambda(A_0) \not\geq \mathrm{D}\Lambda_C^+$.

Case 2: We use Proposition 5.5 together with the facts that $A_D - A_0$ is negative semidefinite in U and $A_D - A_0 \leq -cI$ in B for some scalar $c > 0$, which results in

$$\begin{aligned}
&\langle f_j, \left[\Lambda(A_0) - \Lambda(A_D) + \mathrm{D}\Lambda_C^-\right] f_j\rangle \\
&\qquad \leq \int_\Omega (A_D - A_0)\nabla u_j \cdot \overline{\nabla u_j}\,\mathrm{d}x + \int_C \left(\tfrac{\beta^2}{\alpha} I - A_0\right)\nabla u_j \cdot \overline{\nabla u_j}\,\mathrm{d}x \\
&\qquad \leq \|A_D - A_0\|_* \int_{\Omega\setminus U} |\nabla u_j|^2\,\mathrm{d}x - c\int_B |\nabla u_j|^2\,\mathrm{d}x + \|\tfrac{\beta^2}{\alpha} I - A_0\|_* \int_C |\nabla u_j|^2\,\mathrm{d}x.
\end{aligned}$$

Since $C \subset \Omega \setminus U$, we conclude that

$$\lim_{j\to\infty} \langle f_j, \left[\Lambda(A_0) - \Lambda(A_D) + \mathrm{D}\Lambda_C^-\right] f_j\rangle = -\infty,$$

in particular, $\mathrm{D}\Lambda_C^- \not\geq \Lambda(A_D) - \Lambda(A_0)$.

9. Proof of Theorem 4.2

Assume $B \subseteq D$ and let $u_0 = u_f^{A_0}$. By assumption, $A_D - A_0 \geq cI$ in D for some $c > 0$, from which Proposition 5.6(ii) implies that

$$A_0 A_D^{-1}(A_D - A_0) \geq c\big(\tfrac{\alpha}{\beta}\big)^2 I \text{ in } D.$$

By Proposition 5.5, we have

$$\begin{aligned}\langle f, \big[\mathrm{D}\widehat{\Lambda}_B^{+} + \Lambda(A_0) - \Lambda(A_D)\big] f\rangle &\geq -c\big(\tfrac{\alpha}{\beta}\big)^2 \int_B |\nabla u_0|^2 \,\mathrm{d}x + \int_\Omega A_0 A_D^{-1}(A_D - A_0)\nabla u_0 \cdot \overline{\nabla u_0} \,\mathrm{d}x \\ &\geq c\big(\tfrac{\alpha}{\beta}\big)^2 \int_{D\setminus B} |\nabla u_0|^2 \,\mathrm{d}x \geq 0.\end{aligned}$$

Next, assume $B \not\subseteq D^\bullet$. Since $D^\bullet$ is closed and has connected complement, while B is a non-empty open set, there exist a relatively open and connected set $U \subseteq \overline{\Omega} \setminus D^\bullet$ intersecting Γ and B, and an open ball $\widehat{B} \subset U \cap B$. From Theorem 6.3 there is a sequence (f_j) in $L^2_\diamond(\Gamma)$ such that $u_j = u_{f_j}^{A_0}$ satisfy

$$\lim_{j\to\infty} \int_{\widehat{B}} |\nabla u_j|^2 \,\mathrm{d}x = \infty \qquad \text{and} \qquad \lim_{j\to\infty} \int_{\Omega\setminus U} |\nabla u_j|^2 \,\mathrm{d}x = 0.$$

Since $D \subset \Omega \setminus U$, Proposition 5.5 gives

$$\begin{aligned}\langle f_j, \big[\mathrm{D}\widehat{\Lambda}_B^{+} + \Lambda(A_0) - \Lambda(A_D)\big] f_j\rangle &\leq -c\big(\tfrac{\alpha}{\beta}\big)^2 \int_B |\nabla u_j|^2 \,\mathrm{d}x + \int_\Omega (A_D - A_0)\nabla u_j \cdot \overline{\nabla u_j} \,\mathrm{d}x \\ &\leq -c\big(\tfrac{\alpha}{\beta}\big)^2 \int_{\widehat{B}} |\nabla u_j|^2 \,\mathrm{d}x + \|A_D - A_0\|_* \int_{\Omega\setminus U} |\nabla u_j|^2 \,\mathrm{d}x.\end{aligned}$$

We conclude that

$$\lim_{j\to\infty} \langle f_j, \big[\mathrm{D}\widehat{\Lambda}_B^{+} + \Lambda(A_0) - \Lambda(A_D)\big] f_j\rangle = -\infty,$$

in particular, $\Lambda(A_D) - \Lambda(A_0) \not\leq \mathrm{D}\widehat{\Lambda}_B^{+}$. This proves the contrapositive formulation of the assertion

$$\Lambda(A_D) - \Lambda(A_0) \leq \mathrm{D}\widehat{\Lambda}_B^{+} \quad \text{implies} \quad B \subseteq D^\bullet.$$

10. Proof of Theorem 4.3

Assume $B \subseteq D$ and let $u_0 = u_f^{A_0}$. By assumption, $A_0 - A_D \geq cI$ in D, so Proposition 5.5 yields

$$\begin{aligned}\langle f, \big[\Lambda(A_D) - \Lambda(A_0) - \mathrm{D}\widehat{\Lambda}_B^{-}\big] f\rangle &\geq -c \int_B |\nabla u_0|^2 \,\mathrm{d}x + \int_\Omega (A_0 - A_D)\nabla u_0 \cdot \overline{\nabla u_0} \,\mathrm{d}x \\ &\geq c \int_{D\setminus B} |\nabla u_0|^2 \,\mathrm{d}x \geq 0.\end{aligned}$$

Next, assume $B \not\subseteq D^\bullet$. Since $D^\bullet$ is closed and has connected complement, while B is a non-empty open set, there exist a relatively open and connected set $U \subseteq \overline{\Omega} \setminus D^\bullet$ intersecting Γ and B, and an open ball $\widehat{B} \subset U \cap B$. From Theorem 6.3 there is a sequence (f_j) in $L^2_\diamond(\Gamma)$ such that $u_j = u_{f_j}^{A_0}$ satisfy

$$\lim_{j\to\infty} \int_{\widehat{B}} |\nabla u_j|^2 \,\mathrm{d}x = \infty \qquad \text{and} \qquad \lim_{j\to\infty} \int_{\Omega\setminus U} |\nabla u_j|^2 \,\mathrm{d}x = 0.$$

Since $D \subset \Omega \setminus U$, Proposition 5.5 gives

$$\begin{aligned}\langle f_j, \big[\Lambda(A_D) - \Lambda(A_0) - \mathrm{D}\widehat{\Lambda}_B^{-}\big] f_j\rangle &\leq -c \int_B |\nabla u_j|^2 \,\mathrm{d}x + \int_\Omega A_0 A_D^{-1}(A_0 - A_D)\nabla u_j \cdot \overline{\nabla u_j} \,\mathrm{d}x \\ &\leq -c \int_{\widehat{B}} |\nabla u_j|^2 \,\mathrm{d}x + \|A_0 A_D^{-1}(A_0 - A_D)\|_* \int_{\Omega\setminus U} |\nabla u_j|^2 \,\mathrm{d}x.\end{aligned}$$

We conclude that

$$\lim_{j\to\infty} \langle f_j, \big[\Lambda(A_D) - \Lambda(A_0) - \mathrm{D}\widehat{\Lambda}_B^{-}\big] f_j\rangle = -\infty,$$

in particular, $\mathrm{D}\widehat{\Lambda}_B^- \not\leq \Lambda(A_D) - \Lambda(A_0)$. This proves the contrapositive formulation of the assertion

$$\mathrm{D}\widehat{\Lambda}_B^- \leq \Lambda(A_D) - \Lambda(A_0) \quad \text{implies} \quad B \subseteq D^\bullet.$$

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